

Guide: Running and Verifying Mosquitto MQTT Broker on Windows

This document explains how to install, start, verify, and troubleshoot the Mosquitto MQTT broker on a Windows system for ESP32 or IoT projects.

1. Starting Mosquitto as a Windows Service

If Mosquitto was installed using the official Windows installer and the 'Install Service' option was selected, you can start it as a background service.

Commands:

Start Mosquitto service	net start mosquitto
Stop Mosquitto service	net stop mosquitto
Check service status	sc query mosquitto

2. Running Mosquitto Manually (Debug Mode)

You can run Mosquitto manually to view connection logs and debug messages.

Steps:

1. Open Command Prompt or PowerShell as Administrator.
2. Navigate to Mosquitto installation folder: `cd "C:\Program Files\mosquitto"`
3. Start Mosquitto with verbose output: `mosquitto.exe -v`
4. Or specify configuration file explicitly: `mosquitto.exe -c "C:\Program Files\mosquitto\mosquitto.conf" -v`

3. Verifying Mosquitto is Running

Use the following commands to confirm that Mosquitto is active and listening on port 1883.

Check listening ports	netstat -ano find "1883"
Test subscribe (new window)	mosquitto_sub -h 127.0.0.1 -t test/topic -v
Test publish (another window)	mosquitto_pub -h 127.0.0.1 -t test/topic -m "hello world"

4. Troubleshooting

Mosquitto won't start	Run as Administrator or check if another instance is already running.
ESP32 can't connect	Ensure Windows Firewall allows inbound TCP on port 1883.
No connection logs shown	Run 'mosquitto.exe -v' manually for live debugging output.
Connection refused	Check if your configuration file allows 'allow_anonymous true' or add user

Tip: You can create a batch file (e.g., 'start_mosquitto.bat') with the command 'mosquitto.exe -v' to easily start the broker with logs visible.